

Geometric constraints on the Universe from Type Ia supernovae using Bayesian inference

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Type Ia supernovae provide one of the most direct observational probes of the cosmic expansion history through the distance–redshift relation. In this work, the Union 2.1 Supernova Cosmology Project compilation of 714 spectroscopically confirmed Type Ia supernovae ($0.0025 \leq z \leq 1.4140$) is used to constrain cosmological parameters within the Λ CDM framework. Parameter inference is performed using Bayesian posterior sampling with a Markov Chain Monte Carlo (MCMC) method, and is validated using χ^2 minimisation. For a spatially flat model and a flat prior, the MCMC analysis yields $\Omega_\Lambda = 0.708^{+0.018}_{-0.019}$ and $H_0 = 69.5^{+0.3}_{-0.3} \text{ km s}^{-1} \text{ Mpc}^{-1}$ (68% credible intervals). Allowing spatial curvature significantly weakens the constraints for dark energy, giving $\Omega_\Lambda = 0.70^{+0.11}_{-0.11}$, $H_0 = 69.5^{+0.4}_{-0.4} \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_k = 0.013^{+0.181}_{-0.172}$. The results demonstrate the strong parameter degeneracies present in supernova-only cosmology, particularly between the density parameters when curvature is included. The influence of prior assumptions and redshift cuts is explored to investigate how degeneracies may be reduced. While the expansion history is robustly recovered, spatial curvature remains weakly constrained without external information and high redshift data. These results highlight both the power and the limitations of Type Ia supernovae as cosmological probes and illustrate the importance of combining multiple observational datasets to break geometric degeneracies.

I. INTRODUCTION

A. Observational Cosmology

Modern cosmology seeks to determine the expansion history, composition and geometry of the Universe through observational measurements. A key quantity is the relation between distance and redshift, which encodes the dynamics of cosmic expansion and therefore constrains parameters describing the Universe’s composition. Several independent observational probes contribute to this effort, including measurements of the cosmic microwave background (CMB), baryon acoustic oscillations (BAO) and galaxy clustering and gravitational lensing [1].

Among these probes, Type Ia supernovae play a particularly important role because they provide direct measurements of cosmological distances at redshifts up to $z \sim 1$. These events are believed to arise from thermonuclear explosions of white dwarfs in binary systems and after empirical standardisation corrections, they exhibit remarkably uniform peak luminosities [2]. As a result, they can be treated as standard candles with an observed brightness that relates directly to their luminosity distance.

Observations of distant Type Ia supernovae in the late 1990s revealed that the expansion of the Universe is accelerating, implying the existence of dark energy [3, 4]. Within the standard cosmological model, this component is described by the cosmological constant Λ , which contributes a constant energy density characterised by the parameter Ω_Λ . The physical nature of dark energy remains unknown as it does not interact directly with electromagnetic radiation but its presence can be inferred by its gravitational influence. Another parameter describing the composition of the Universe is the spatial curvature, Ω_k , which characterises the geometry of spacetime. If $\Omega_k > 0$ the Universe has an open geometry with negative curvature, if $\Omega_k < 0$ the Universe is closed with positive curvature, and if $\Omega_k = 0$ the Universe is spatially flat.

The Union 2.1 dataset produced by the Supernova Cosmology Project [5] combines several subsequent supernovae surveys into a homogenised catalogue of spectroscopically confirmed Type Ia supernovae. Analysing the distance–redshift relation of this dataset provides a direct

way to test the Λ CDM model and investigate the degeneracies that arise between cosmological parameters.

The interpretation of supernova observations requires careful statistical inference. As multiple combinations of parameters can produce similar expansion histories, Bayesian methods are well suited to quantifying the parameter uncertainties as they explore the posterior probability distributions.

B. Theory

The luminosity distance measured for supernovae, and therefore the predicted distance modulus, depends on the expansion history of the Universe. The theoretical framework used in this analysis follows the standard cosmological model described in Liddle (2015).

The density parameters are related as,

$$1 = \Omega_M + \Omega_\Lambda + \Omega_k + \Omega_R, \quad (1)$$

where Ω_M , Ω_Λ , Ω_k and Ω_R are the density parameters for matter, dark energy, curvature and radiation respectively. This project neglects the radiation density parameter as its present day value is close to zero. The expansion dynamics are governed by the Friedmann equation:

$$H(z)^2 = H_0^2 [\Omega_M(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda], \quad (2)$$

where $H(z)$ is the Hubble parameter at redshift z and H_0 is its present day value. This can be rewritten without the dependence on the matter density parameter by substituting $\Omega_M = 1 - \Omega_\Lambda - \Omega_k$.

The comoving distance to a source at redshift z is given by,

$$r_c(z) = c \int_0^z \frac{dz'}{H(z')} \quad (3)$$

and the transverse comoving distance introduces geometric corrections for curved space:

$$S_k(z) = \begin{cases} \frac{c}{H_0 \sqrt{\Omega_k}} \sinh \frac{\sqrt{\Omega_k} H_0 r_c}{c} & \text{if } \Omega_k > 0 \\ r_c(z) & \text{if } \Omega_k = 0 \\ \frac{c}{H_0 \sqrt{|\Omega_k|}} \sin \frac{\sqrt{|\Omega_k|} H_0 r_c}{c} & \text{if } \Omega_k < 0. \end{cases} \quad (4)$$

The speed of light, c , is set at $299792458.0 \text{ m s}^{-1}$ for this project [7]. Finally, the luminosity distance is

$$D_L(z) = (1+z)S_k(z), \quad (5)$$

which relates to the observed distance modulus

$$\mu(z) = 5 \log_{10} \frac{D_L}{pc} - 5. \quad (6)$$

Multiple combinations of Ω_Λ , H_0 and Ω_k can produce very similar luminosity distances, causing degeneracies to naturally arise.

The remainder of this report is structured as follows. Section II describes the supernova dataset, likelihood construction, the Markov Chain Monte Carlo sampling procedure as well as a summary of the various inference choices that are explored. Section III presents the resulting parameter constraints for flat and curved cosmological models and summarises the convergence tests performed. Following this, three main investigations into inference choices are explored: various dimensions to expose degeneracies, different priors to explore 'breaking' a degeneracy and redshift cuts to target curvature constraint dependencies. Throughout the report, the primary results are presented visually through figures to maintain clarity in the main discussion. Numerical values for fitted parameters and additional details of the priors and convergence diagnostics are provided in the appendices to preserve full reproducibility of the analysis.

II. DATA AND METHODOLOGY

A. Supernovae Data

This analysis uses the Union 2.1 compilation from the Supernova Cosmology Project as presented in Suzuki et al. (2012), containing spectroscopically confirmed Type Ia supernovae spanning redshifts $0.0025 \leq z \leq 1.4140$. A representative subset of the data is reproduced in Appendix V. 714 of the supernovae had the relevant data: measured distance modulus μ_i , an associated observational uncertainty $\sigma_{\mu,i}$ and the corresponding redshift z_i . Throughout this report, parameters from Aghanim et al. (2020) are used as the comparative literature data.

The quoted uncertainties are reported directly in magnitude, and since magnitude is related logarithmically to flux through $\mu = -2.5 \log_{10} F + \text{const}$, small flux uncertainties propagate approximately linearly into magnitude uncertainties. As the observational errors (photometric noise, intrinsic scatter) are largely independent, the resulting magnitude errors are expected to be approximately Gaussian distributed, hence motivating the use of a Gaussian likelihood. Uncertainties in redshift are typically negligible compared to magnitude uncertainties and are therefore neglected in this report's analysis.

The comoving distance integral is computed using the trapezoidal rule on a grid of 2000 equally spaced points up to the maximum redshift $z_{max} = 1.4140$, corresponding to a spacing $\Delta z \approx 7 \times 10^{-4}$. Considering that $H(z)$ is smooth, the interpolation between grid points introduces a numerical error that is negligible compared to the observational magnitude uncertainties (typically $\sim 0.1 \text{ mag}$).

B. Likelihood Function and χ^2 Implementation

Assuming independent Gaussian errors in the observed distance moduli, the likelihood for a parameter vector $\theta = (\Omega_\Lambda, H_0, \Omega_k)$ can be written as

$$L(\theta) \propto \exp \left(-\frac{\chi^2(\theta)}{2} \right), \quad (7)$$

where the χ^2 statistic is

$$\chi^2(\theta) = \sum_i \frac{[\mu_i - \mu_{\text{model}}(z_i; \theta)]^2}{\sigma_{\mu,i}^2}. \quad (8)$$

Minimising χ^2 is equivalent to maximising the Gaussian likelihood. Under the assumption of a locally quadratic likelihood surface, χ^2 minimisation provides parameter estimates and approximate uncertainties from curvature around the minimum. However, this approach implicitly assumes Gaussian-distributed residuals, a nearly Gaussian posterior and weak parameter degeneracies. When degeneracies are strong or posteriors are non-elliptical, χ^2 -based error estimates can become unreliable. For this reason, χ^2 minimisation is used primarily as a validation tool and for comparison with the full Bayesian posterior sampling [8].

C. MCMC Sampling and Posterior Inference

Bayesian inference is performed using the affine-invariant ensemble sampler implemented in the emcee package [9]. The posterior is defined as

$$P(\theta|data) \propto L(\theta)\pi(\theta) \quad (9)$$

where $\pi(\theta)$ denotes the prior. The walkers were initialised near fiducial Λ CDM values, $(\Omega_\Lambda, H_0, \Omega_k)_{init} = (0.7, 70 \text{ km s}^{-1} \text{ Mpc}^{-1}, 0)$, with small Gaussian perturbations. The number of walkers and steps were increased with dimension to ensure reasonable autocorrelation times, and one fifth of the total steps were discarded as burn in. Convergence of the MCMC chains was assessed using autocorrelation times, acceptance fractions, and visual inspection of walker trajectories. A summary of these diagnostics is presented in Section III B, while additional convergence statistics and a walker trace plot are provided in Appendix V.

Unlike χ^2 minimisation, MCMC samples the full posterior distribution rather than assuming Gaussianity and naturally captures curved or elongated degeneracy directions. The results provide asymmetric credible intervals, 16th-84th percentiles of the marginal posterior distribution, as well as covariance matrices and eigenvectors describing parameter correlations. This is particularly important in higher dimensional models, where degeneracies between parameters cause non-trivial posterior shapes. Agreement between χ^2 best-fit values and posterior medians serves as a consistency check when the posterior is approximately Gaussian.

Markov Chain Monte Carlo methods are widely used in cosmology to explore high-dimensional parameter spaces; for example, the Aghanim et al. (2020) analysis used MCMC sampling to constrain cosmological parameters from the cosmic microwave background data.

D. Models and Inference Choices

To assess the robustness of parameter constraints and the impact of model assumptions, several cosmological and inference configurations are explored.

Three Λ CDM parameterisations are considered:

- 1D: Ω_Λ free, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_k = 0$
- 2D: Ω_Λ and H_0 free, $\Omega_k = 0$
- 3D: Ω_Λ , H_0 and Ω_k free

The 1D configuration isolates sensitivity to dark energy under strong geometric assumptions. The 2D model exposes the degeneracy between Ω_Λ and H_0 , while the 3D case targets the weak supernova dependency on Ω_k and demonstrates its impact on other parameter constraints.

Parameter inference is repeated under several prior prescriptions in the Ω_Λ - H_0 space for the 2D model to evaluate prior sensitivity. These include:

- Broad uniform (“flat wide”) priors,
- Restricted uniform (“flat narrow”) priors,
- Gaussian priors motivated by CMB constraints (“CMB Gaussian”),
- Correlated CMB-informed priors aligned with the principal degeneracy direction (“CMB directional”).

The numerical quantifications of the priors are summarised separately in Appendix V. Both flat priors are built from reasonable bounds for each parameter given the literature from Aghanim et al. (2020). The CMB Gaussian prior takes the literature values for Ω_Λ and H_0 , inflates the quoted errors by a factor of five and builds a 2D Gaussian surface. The CMB directional prior required digitising Fig.17 from the Aghanim et al. report to provide a covariance matrix between Ω_Λ and H_0 . Again, the uncertainty was inflated by a factor of five to ensure that the prior did not dominate the supernovae likelihood. Comparing posteriors under different priors allows assessment of the information required to successfully reduce degeneracies and improve parameter constraints from supernovae data.

Additional tests were performed using redshift cuts to examine sensitivity to low- and high-redshift supernovae for the 3D model. Since curvature effects scale as $(1+z)^2$, it is expected that high-redshift events are important for constraining Ω_k , while low-redshift supernovae primarily anchor Ω_Λ and H_0 .

These tests help identify the most effective analysis or measurement techniques to improve cosmological parameter constraints from Type Ia supernovae data. Details of parameter uncertainty estimations are described in Appendix V.

III. RESULTS AND DISCUSSION

A. Fit to the Distance-Redshift Relation

Figure 1 shows the best-fitting curved model obtained with the flat wide prior MCMC procedure, overlaid on the supernova distance modulus–redshift data. The corresponding normalised residuals are shown in Figure 2 for both flat

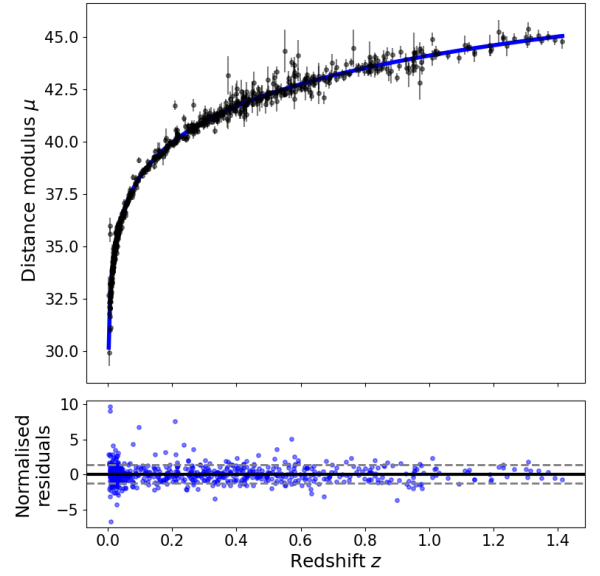


FIG. 1: Distance modulus-redshift relation with data points in black and best-fit curve in blue for a curved Universe model. Error bars for the Union 2.1 data are displayed in the y-direction (distance modulus) whilst no errors were reported in redshift. Normalised residuals are displayed underneath in blue, with a black line indicating a value of 0 and the grey dashed lines indicating values ± 1 .

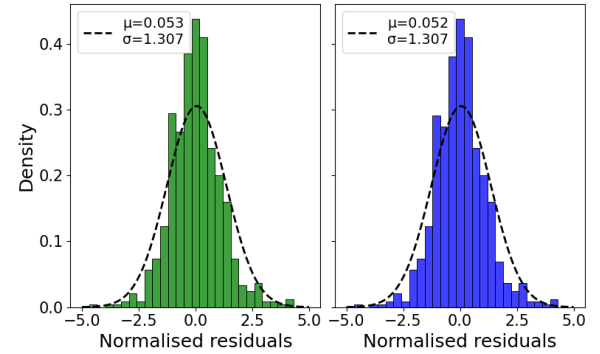


FIG. 2: Normalised residual histograms for both a flat Universe model in green and a curved Universe in blue. Both feature a dashed line indicating the Gaussian fit to the histogram with the best-fit means and standard errors quoted to three decimal places in the legends.

and curved geometries. The flat model returned parameters $\Omega_\Lambda = 0.708^{+0.018}_{-0.019}$ and $H_0 = 69.5^{+0.3}_{-0.3} \text{ km s}^{-1} \text{ Mpc}^{-1}$. The curved model returned values of $\Omega_\Lambda = 0.70^{+0.11}_{-0.11}$, $H_0 = 69.5^{+0.4}_{-0.4} \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_k = 0.01^{+0.18}_{-0.17}$.

The two best-fit curves are indistinguishable within the scatter of the data. The mean normalised residual changes only marginally from 0.052 (curved) to 0.053 (flat), while the standard deviation remains the same at 1.31 to three significant figures for both models. In each case, the residuals are symmetrically distributed about zero and exhibit no significant redshift-dependent structure, indicating that the models successfully reproduce the global behaviour of the dataset. The measured residual scatter of $\sigma_r = 1.31$ and the reduced $\chi^2 = 1.72$ for both models correspond as expected through the relation $\chi^2_\nu \approx \sigma_r^2$.

Since the residuals are normalised by the quoted observational uncertainties, a standard deviation greater than

unity indicates that the total dispersion exceeds that expected from measurement errors alone. This excess variance reflects additional dispersion not fully captured by the reported magnitude errors. In supernova cosmology, this is commonly attributed to the intrinsic luminosity dispersion of Type Ia supernovae together with small calibration and systematic uncertainties. The elevated scatter therefore reflects limitations in the noise model rather than a breakdown of the cosmological description.

The near-identical goodness-of-fit between flat and curved models provides an early indication of parameter degeneracy. Curvature therefore cannot be distinguished by inspection of the Hubble diagram alone, motivating a more detailed analysis of parameter constraints.

B. Convergence Diagnostics

Before interpreting parameter constraints, the reliability of the inference must be established. As advised in Foreman-Mackey et al. (2013), convergence was assessed using the integrated autocorrelation time, which can be used to estimate the effective number of independent samples in the chain. Convergence diagnostics were examined for all model dimensions using the flat wide prior. MCMC settings were increased from 50 walkers at 2000 steps for the 1D case up to 100 walkers and 5000 steps for the 3D case.

Integrated autocorrelation times ranged from 23.0 to 36.8 steps, with the increase reflecting the growing complexity of the parameter space at higher dimensions. In all cases, the total chain lengths exceed several times the autocorrelation time, implying many independent samples and ensuring adequate sampling of the posterior distributions. For the 2D and 3D cases, the mean acceptance fractions lie within the expected range (0.2–0.8) for affine-invariant samplers, indicating efficient exploration of parameter space. The 1D case has fewer correlations to explore and hence the MAF is slightly higher than the expected range. Full convergence statistics and interpretations are discussed in Appendix V.

As an additional test of the robustness, the spacing of the walker initialisations was varied and the change in the fitted parameters analysed. By both doubling and halving the initial walker spread, the posterior medians and error bar widths changed by less than 1% for both Ω_Λ and H_0 . There was greater variation in the results for Ω_k , with the median changing by 26.3% when the spacing was halved and by 8.88% when doubled. The corresponding 16th–84th percentile widths changed by 1.08% and 0.766%. This demonstrates that the results for Ω_Λ and H_0 are stable within the chosen MCMC settings, although Ω_k is more sensitive to the inference choices, reflecting the weaker constraints that supernova data place on spatial curvature.

C. Evolution Through Dimensions

Figure 3 summarises the evolution of cosmological parameter constraints for Ω_Λ and H_0 as the model dimensionality increases from 1D (Ω_Λ only) to 2D (Ω_Λ , H_0) to 3D (Ω_Λ , H_0 , Ω_k). The figure compares posterior medians obtained from MCMC sampling with best-fit values derived from χ^2 minimisation, as well as the literature results. The data and corner plots for the results shown in Figure 3 can be

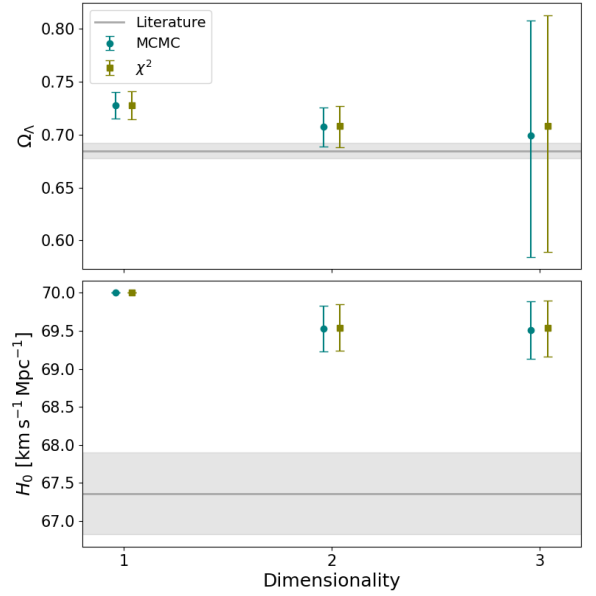


FIG. 3: Evolution of Ω_Λ and H_0 with increasing dimensionality. Fitted parameters and error bars are displayed at each dimension for both MCMC and χ^2 analysis in cyan and olive respectively. Literature values and their error bars are shown as grey bands for each quantity.

found in Appendix V. Across all dimensions, strong agreement is observed between the χ^2 best-fits and the MCMC posterior medians, with consistent uncertainty estimates, hence validating the implementation of the MCMC code.

In the 1D model, Ω_Λ is tightly constrained due to the assumption of fixed H_0 and spatial flatness. When H_0 is introduced as a free parameter in the 2D model, the posterior distribution of Ω_Λ broadens, producing elongated contours and increasing the error bar range by 47.9%. This infers that different combinations of H_0 and Ω_Λ generate nearly identical luminosity distance curves.

This degeneracy effect is significantly more pronounced in the 3D model when curvature is allowed to vary. The inclusion of Ω_k introduces additional geometric freedom, causing the error bar range for Ω_Λ to increase again by a further 500%. The corresponding corner plots display extended, tilted contours characteristic of strong degeneracies.

Comparison with results from Aghanim et al. (2020) shows good agreement for Ω_Λ within uncertainties. However, a clear discrepancy is observed for H_0 , consistent with the well-documented Hubble tension between early-Universe measurements and late-Universe probes such as supernovae. While resolving this tension lies beyond the scope of this project, its appearance here confirms that the analysis reproduces established features of contemporary cosmological datasets [10].

The growth of parameter degeneracy with increasing model dimensionality is quantified using the covariance matrix eigenvalues (λ) and the condition number

$$C = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Higher condition numbers imply higher sensitivity to perturbations, and hence describes a lack of numerical stability due to degeneracies. In the 1D model, the variance is $\sigma^2 = 1.62 \times 10^{-4}$ indicating tight constraints when only Ω_Λ is free.

In the 2D flat model, the covariance matrix eigenvalues span $\lambda_{\min} = 1.72 \times 10^{-4}$ and $\lambda_{\max} = 9.30 \times 10^{-2}$, giving a condition number $C = 5.41 \times 10^2$. This reflects a pronounced degeneracy between Ω_Λ and H_0 , with one direction in parameter space weakly constrained.

In the full 3D model allowing curvature, the eigenvalues broaden further: $\lambda_{\min} = 1.31 \times 10^{-4}$ and $\lambda_{\max} = 1.64 \times 10^{-1}$ yielding $C = 1.25 \times 10^3$. The near-doubling of the condition number confirms that introducing Ω_k substantially increases degeneracy. There is a negative correlation between Ω_Λ and Ω_k due to the relation $\Omega_M = 1 - \Omega_\Lambda - \Omega_k$.

These results quantitatively confirm that curvature remains weakly constrained by supernova data alone, and its degeneracy with Ω_Λ reduces the precision on the dark energy density parameter. Overall, increasing model dimensionality primarily introduces parameter degeneracies, significantly weakening individual constraints. This behaviour highlights a fundamental limitation of supernova-only cosmology in multi-parameter models.

D. Influence of Priors

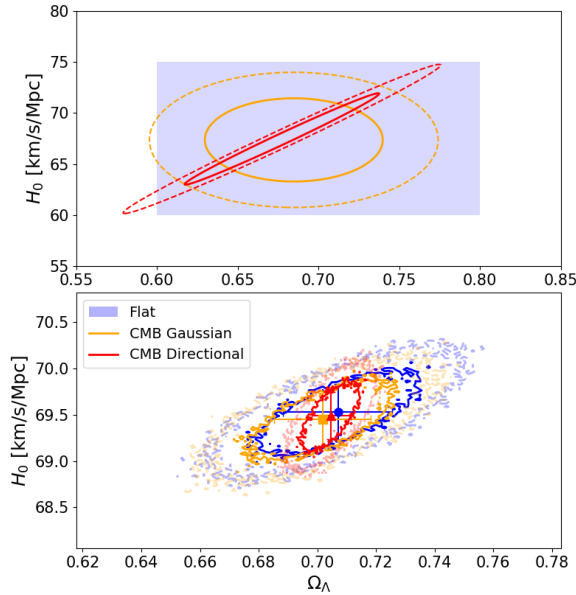


FIG. 4: Flat narrow, CMB Gaussian and CMB directional priors visualised in $\Omega_\Lambda - H_0$ space in the top panel along with their resulting 68% and 95% percentile contours of the posterior distributions for MCMC sampling in the lower panel. Matching colours are used across both panels with flat narrow in blue, CMB Gaussian in yellow and CMB directional in red. The visualisation for both CMB priors show a 68% contour in a solid line and a 95% contour in a dashed line. Posterior medians along with error bars in both directions are shown for each prior in the lower panel.

The impact of different prior choices on parameter inference was explored using both supernova-only and CMB-based [1] priors for a flat universe. Figure 4 visualises three of the priors used: flat narrow, CMB Gaussian, and CMB directional along with the resulting 68% and 95% posterior contours for Ω_Λ and H_0 . A summary of the fitted parameters and their errors is shown in Table I.

The flat narrow and CMB Gaussian priors generate posterior contours of similar size and orientation, although the Gaussian prior pulls the central value for both parameters to

	Ω_Λ	H_0 (km s $^{-1}$ Mpc $^{-1}$)
Flat Narrow	$0.707^{+0.018}_{-0.019}$	$69.5^{+0.3}_{-0.3}$
CMB Gaussian	$0.702^{+0.016}_{-0.017}$	$69.5^{+0.3}_{-0.3}$
CMB Directional	$0.705^{+0.007}_{-0.007}$	$69.5^{+0.3}_{-0.3}$
Planck 2018	0.6847 ± 0.0073	67.36 ± 0.54

TABLE I: Medians and 16th-84th percentile error bars for Ω_Λ and H_0 parameters from flat narrow, CMB Gaussian and CMB directional informed posteriors as well as the literature data as quoted in Aghanim et al. (2020).

smaller values, consistent with the known Hubble tension. In contrast, the CMB directional prior produces a substantially smaller contour, rotated relative to the others.

While the posterior uncertainties on H_0 remain similar across all priors (around ± 0.3 km s $^{-1}$ Mpc $^{-1}$), the 68% credible interval for Ω_Λ only approaches the precision of the literature when the CMB directional prior is applied. This causes the error to reduce from ± 0.018 (flat narrow) to ± 0.007 with the directional prior, closely matching CMB's ± 0.0073 . This illustrates that significant improvement on degenerate parameter constraints can only be achieved with information on external degeneracy directions. Simply applying priors related to other parameter fits does not suffice to break the intrinsic supernova-only degeneracy.

E. Redshift Slicing

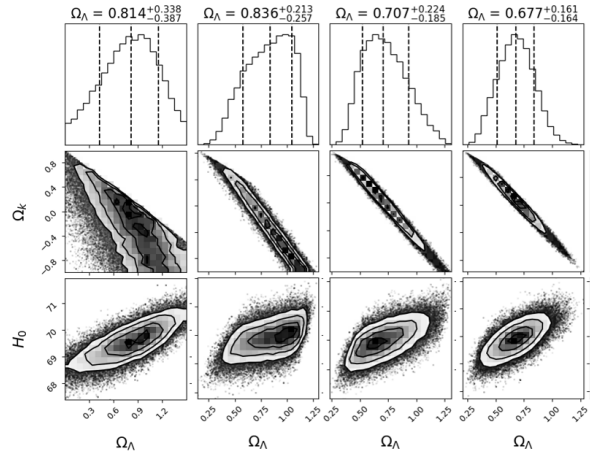


FIG. 5: Cropped corner plots for three-dimensional MCMC fitting across various redshift cuts of the supernovae data. Focused panels correspond to the Ω_Λ contours with the other two parameters, H_0 and Ω_k , as well as the histogram of the Ω_Λ posterior distribution. The displayed redshift cuts are $z \leq 0.1$, $z \leq 0.36$, $z \leq 0.63$ and $z \leq 0.89$ from left to right.

To investigate the dependence of constraining power on redshift, MCMC fits were performed on subsets of the data defined by maximum redshift cuts: $z \leq 0.1$, $z \leq 0.36$, $z \leq 0.63$, $z \leq 0.89$, $z \leq 1.15$, and $z \leq 1.41$. The cropped corresponding posterior contours for several selected redshift cuts are shown in Figure 5. Posterior medians and error bars for each redshift cut are shown in Appendix V.

At low redshift ($z \leq 0.1$), the posterior contours are non-elliptical, and the distribution of Ω_Λ is strongly non-Gaussian. This reflects the limited geometric leverage provided by nearby supernovae. As higher-redshift supernovae

are included, the contours become increasingly elliptical and more Gaussian in shape. The boundary in the $z \leq 0.10$ contour is due to the requirement that Ω_M is positive.

Examining the evolution of parameter uncertainties with increasing redshift shows that the credible intervals for parameters decrease systematically. Over the six redshift cuts, the error bar range decreases by 27.7% on average for Ω_Λ and by 24.6% for Ω_k . The results are more stable for H_0 , with the uncertainty reducing from ± 0.599 at $z \leq 0.10$ to $\pm 0.380 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for the full redshift range.

High-redshift supernovae provide increased leverage on spatial curvature, which in turn impacts the inferred dark energy density through the strong degeneracy between these parameters. However, without prior knowledge from the Λ CDM model, the results could equally be interpreted in the opposite direction, with variations in Ω_Λ compensating for curvature. This behaviour follows from Equation 2, where curvature contributions grow as $(1+z)^2$ while the dark energy term is redshift independent. Although both parameters appear in the $(1+z)^3$ term through the substitution for Ω_M , this contribution affects them the same way.

For all redshift cuts up to the full distribution ($z \leq 1.41$), the median Ω_Λ remains consistent with the literature within error, demonstrating that while supernovae robustly probe the expansion history, it is only the parameter constraints that are heavily affected by parameter degeneracies, not their accuracies.

F. Summary of Investigations

Taken together, the three investigations illustrate a coherent picture of how supernova-only constraints are shaped by both data and prior information. Increasing model dimensionality introduces strong parameter degeneracies as observed by broadened posteriors and inflated condition numbers, while external priors aligned with degeneracy directions can selectively tighten constraints, particularly on Ω_Λ . Similarly, extending the redshift range systematically reduces uncertainties, with high-redshift supernovae providing crucial leverage on curvature. These results demonstrate that the precision of supernova cosmology is jointly determined by the interplay of dataset depth, dimensionality, and prior knowledge, highlighting that breaking geometric degeneracies requires both high-redshift coverage and complementary external constraints.

IV. CONCLUSION

This work has used the Union 2.1 Type Ia supernova compilation to constrain cosmological parameters within the Λ CDM framework using Bayesian MCMC inference. For a spatially flat model the results give $\Omega_\Lambda = 0.708^{+0.018}_{-0.019}$ and $H_0 = 69.5^{+0.3}_{-0.3} \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with values reported in the literature. Allowing spatial curvature significantly weakens these constraints, yielding $\Omega_k = 0.013^{+0.181}_{-0.172}$ and demonstrating the well-known geometric degeneracy present in supernova-only analyses.

Exploration of model dimensionality, prior choices and redshift cuts shows that the primary limitation of supernova cosmology is not the recovery of the expansion history itself but the strong correlations between cosmological parameters. High-redshift supernovae provide the most lever-

age on curvature, while external information aligned with degeneracy directions is required to tighten parameter constraints. While Type Ia supernovae provide robust measurements of the distance–redshift relation, breaking geometric degeneracies ultimately requires the combination of multiple independent cosmological datasets such as the cosmic microwave background and baryon acoustic oscillations.

Currently, Type Ia supernovae can only be measured to around $z \approx 2$. As astronomy instrumentation improves and supernovae are measured at higher redshifts, the value of Ω_k will be constrained more tightly. Future work with higher-quality datasets could further extend this analysis by incorporating multiple observational constraints into the likelihood. Further improvements could also include refining the numerical sampling strategy or increasing the number of walkers to better characterise the posterior distributions. Continually improving our precision of the Universe’s fundamental parameters will provide deeper insight into its ultimate evolution and fate.

V. ACKNOWLEDGEMENTS AND GAI STATEMENT

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ERROR APPENDIX

This appendix summarises the statistical interpretation of the χ^2 statistic and the procedures used to estimate parameter uncertainties using both χ^2 analysis and Markov Chain Monte Carlo (MCMC) sampling. The discussion follows the treatment of uncertainty analysis described by Hughes et al. (2010).

A1 χ^2 Goodness-of-Fit

The goodness-of-fit of each cosmological model is evaluated using the minimum χ^2 value obtained from Equation 8. For N data points and p fitted parameters, the number of degrees of freedom is

$$\nu = N - p. \quad (10)$$

The reduced χ^2 is then defined as

$$\chi_\nu^2 = \frac{\chi_{\min}^2}{\nu}. \quad (11)$$

If the model is correct and the quoted uncertainties accurately describe the data, χ_ν^2 is expected to be approximately unity. For large ν , statistical fluctuations around $\chi_{\min}^2$ follow the approximate range

$$\nu - 2\sqrt{2\nu} < \chi_{\min}^2 < \nu + 2\sqrt{2\nu}. \quad (12)$$

For the present dataset with $\nu \approx 700$, this corresponds to an expected χ_ν^2 range of approximately 1 ± 0.1 . The standard deviation of the normalised residuals is defined as

$$r_i = \frac{\mu_i - \mu_{\text{model},i}}{\sigma_{\mu,i}}. \quad (13)$$

The two models explored in Section III A both resulted in $\chi_\nu^2 \approx 1.72$ and $\sigma_r \approx 1.31$, hence satisfying

$$\chi_\nu^2 \approx \sigma_r^2. \quad (14)$$

Discussion in Section III A identifies that these quantities exceed that predicted by the quoted observational uncertainties alone. This elevated residual scatter is consistent with the known intrinsic dispersion of Type Ia supernova luminosities and small systematic uncertainties that are not fully captured in the reported magnitude errors.

A2 Parameter Uncertainties from χ^2

Parameter uncertainties in the χ^2 framework are estimated using the change in χ^2 relative to the minimum value,

$$\Delta\chi^2 = \chi^2 - \chi_{\min}^2. \quad (15)$$

Under the assumption that the likelihood surface is locally quadratic near the minimum, the $\Delta\chi^2$ quantity can be used to define 68% confidence regions depending on the

number of parameters being estimated simultaneously. The corresponding thresholds are

$$\Delta\chi^2 = \begin{cases} 1.00 & (1 \text{ parameter}) \\ 2.30 & (2 \text{ parameters}) \\ 3.53 & (3 \text{ parameters}). \end{cases} \quad (16)$$

In practice, one parameter is varied while the others are re-optimised to determine the points where the target $\Delta\chi^2$ value is reached, yielding asymmetric uncertainty estimates.

A3 Parameter Uncertainties from MCMC

In the Bayesian framework, parameter uncertainties are obtained directly from the posterior distributions generated by the MCMC sampler. Credible intervals are defined using the marginal posterior percentiles, with the 16th and 84th percentiles corresponding to the central 68% credible interval.

For approximately Gaussian posteriors these intervals correspond closely to the $\Delta\chi^2$ confidence intervals specified in Equation 16. However, MCMC methods remain valid even when the posterior distribution is non-Gaussian or strongly degenerate, since they sample the full probability distribution rather than relying on a quadratic approximation to the likelihood.

Agreement between the χ^2 best-fit parameters and the MCMC posterior medians therefore provides an important consistency check for the inference procedure used in this work.

CONVERGENCE CHECKS

	Walkers	Steps	MAF	Condition number	τ_{av}
1D	50	2000	0.808	-	23.0
2D	75	3000	0.715	541	27.8
3D	100	5000	0.646	1250	35.8

TABLE II: Chosen MCMC settings and convergence diagnostics for each model dimension using a flat wide prior. Mean acceptance fractions, condition numbers and autocorrelation times are quoted to three significant figures. Autocorrelation time is quoted as the average over all parameters.

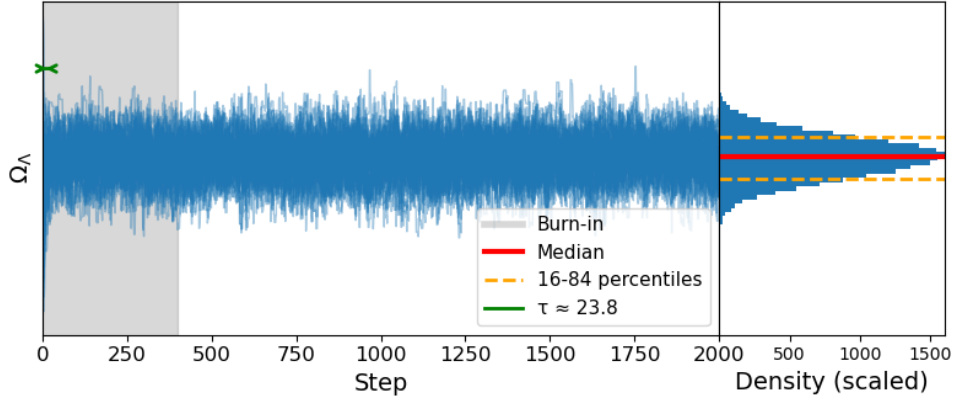


FIG. 6: 1D model walkers plot, highlighting the burn in range and autocorrelation length along with the posterior distribution.

The convergence of the Markov Chain Monte Carlo (MCMC) sampling was assessed using several standard diagnostics, summarised in Table II. In particular, the integrated autocorrelation time τ was used as the primary indicator of chain mixing. The autocorrelation time measures the number of sampling steps required for successive samples in the chain to become effectively independent. Shorter autocorrelation times therefore indicate more efficient sampling and better exploration of the parameter space.

For reliable posterior estimation, the total chain length should significantly exceed the autocorrelation time. In all models considered here the number of sampling steps is substantially larger than the estimated autocorrelation time, indicating that the chains contain many independent samples. The walker evolution shown in Figure 6 further demonstrates that the ensemble quickly disperses through parameter space after an initial burn-in period, after which the walkers oscillate around the high-probability region. This behaviour is consistent with good mixing of the ensemble sampler and stable posterior estimates.

EXTRA DATA AND FIGURES

This appendix provides supplementary plots, parameter tables and details of the prior constructions used in the analysis, along with a subset of the Union 2.1 dataset.

Additional Corner Plots

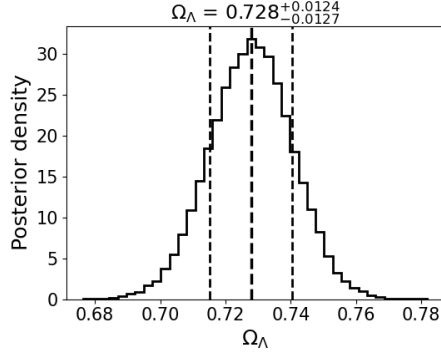


FIG. 7: Corner plot for 1D model under a flat wide prior.

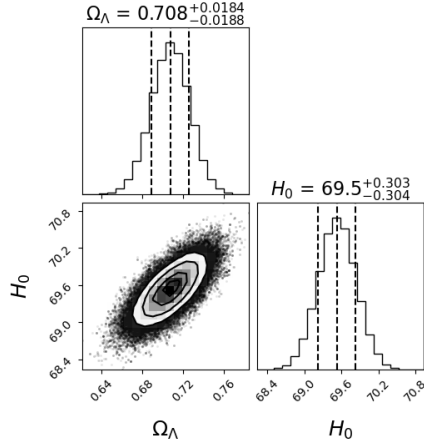


FIG. 8: Corner plot for 2D model under a flat wide prior.

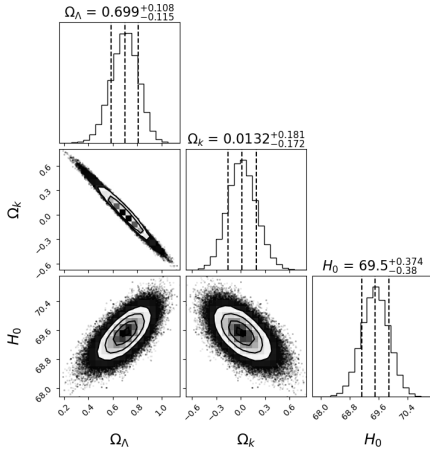


FIG. 9: Corner plot for 3D model under a flat wide prior.

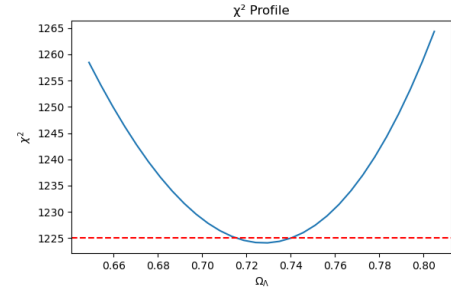


FIG. 10: Corner-like plot for χ^2 minimisation results for 1D model.

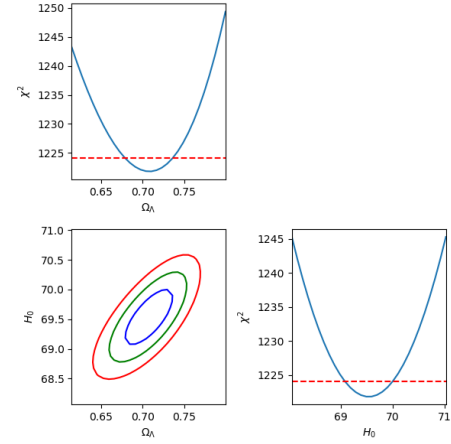


FIG. 11: Corner-like plot for χ^2 minimisation results for 2D model.

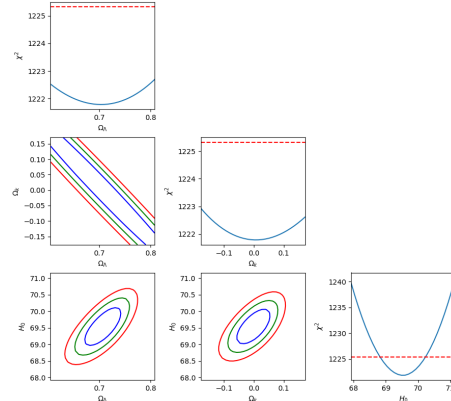


FIG. 12: Corner-like plot for χ^2 minimisation results for 3D model.

Additional Paramater Tables

Values of Ω_k are quoted to three significant figures because the posterior uncertainties are large relative to the median value. Note that the Hubble constant (H_0) units are $\text{km s}^{-1} \text{Mpc}^{-1}$.

MCMC	Ω_Λ	H_0	Ω_k	χ^2	Ω_Λ	H_0	Ω_k
1D	$0.728^{+0.012}_{-0.013}$	-	-	1D	$0.728^{+0.013}_{-0.013}$	-	-
2D	$0.708^{+0.018}_{-0.019}$	$69.5^{+0.3}_{-0.3}$	-	2D	$0.71^{+0.02}_{-0.02}$	$69.5^{+0.3}_{-0.3}$	-
3D	$0.70^{+0.11}_{-0.11}$	$69.5^{+0.4}_{-0.4}$	$0.013^{+0.181}_{-0.172}$	3D	$0.71^{+0.11}_{-0.12}$	$69.5^{+0.4}_{-0.4}$	$0.000^{+0.189}_{-0.166}$

TABLE III: Fitted parameters and their errors for Ω_Λ , H_0 and Ω_k parameters from 1D, 2D and 3D models with a flat wide prior. Left hand side of the table displays results acquired by MCMC sampling and the right hand side show those acquired by χ^2 minimisation.

	Ω_Λ	H_0	Ω_k
$z \leq 0.10$	$0.8^{+0.3}_{-0.4}$	$69.5^{+0.6}_{-0.6}$	$-0.3^{+0.6}_{-0.5}$
$z \leq 0.36$	$0.8^{+0.2}_{-0.3}$	$69.5^{+0.4}_{-0.4}$	$-0.351^{+0.571}_{-0.451}$
$z \leq 0.63$	$0.7^{+0.2}_{-0.2}$	$69.6^{+0.4}_{-0.4}$	$0.034^{+0.348}_{-0.430}$
$z \leq 0.89$	$0.68^{+0.16}_{-0.16}$	$69.5^{+0.4}_{-0.4}$	$0.069^{+0.284}_{-0.282}$
$z \leq 1.15$	$0.76^{+0.13}_{-0.13}$	$69.6^{+0.4}_{-0.4}$	$-0.106^{+0.220}_{-0.208}$
$z \leq 1.41$	$0.70^{+0.11}_{-0.11}$	$69.5^{+0.4}_{-0.4}$	$0.015^{+0.179}_{-0.170}$

TABLE IV: Medians and 16th-84th percentile error bars for Ω_Λ , H_0 and Ω_k parameters for 3D models with a flat wide prior for various redshift cuts.

Prior Details

Prior Type	Ω_Λ	H_0
Flat Wide	$0 < \Omega_\Lambda < 1.5$	$40 < H_0 < 100$
Flat Narrow	$0.6 < \Omega_\Lambda < 0.8$	$60 < H_0 < 75$

TABLE V: Uniform prior ranges used in the MCMC sampling. Physicality of the matter density parameter $\Omega_M = 1 - \Omega_\Lambda - \Omega_k$ was enforced by requiring $\Omega_M > 0$.

Parameter	Mean	Standard Deviation
H_0	67.36	5×0.54
Ω_Λ	$1 - 0.3153$	5×0.0073

TABLE VI: CMB Gaussian priors motivated by Aghanim et al. (2020) cosmological constraints. The variances were inflated by a factor of five to ensure that the prior provides directional guidance without dominating the supernova likelihood.

The CMB directional prior was constructed by digitising the H_0 - Ω_M confidence contour from Figure 17 of Aghanim et al. (2020), returning the covariance matrix below

$$\Sigma_{(H_0, \Omega_M)} = \begin{pmatrix} \sigma_{H_0}^2 & \sigma_{H_0 \Omega_M} \\ \sigma_{H_0 \Omega_M} & \sigma_{\Omega_M}^2 \end{pmatrix} = \begin{pmatrix} 0.3556 & -4.745 \times 10^{-3} \\ -4.745 \times 10^{-3} & 6.447 \times 10^{-5} \end{pmatrix}. \quad (17)$$

Assuming a flat universe ($\Omega_k = 0$), the density parameters satisfy

$$\Omega_\Lambda = 1 - \Omega_M. \quad (18)$$

The covariance matrix derived from the CMB constraint contour was therefore transformed from the parameter space (H_0, Ω_M) to (H_0, Ω_Λ) . This transformation was performed using the Jacobian

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (19)$$

such that

$$\Sigma_{(H_0, \Omega_\Lambda)} = J \Sigma_{(H_0, \Omega_M)} J^T. \quad (20)$$

To prevent the CMB directional prior from dominating the supernova likelihood, the uncertainties were broadened by a factor of five. This widens the allowed parameter region while maintaining the degeneracy direction implied by the CMB constraint.

Example dataset

A small excerpt from the Union 2.1 Type Ia supernova compilation is shown below to illustrate the structure of the dataset used in the analysis.

SN Name	z	m_B	x_1 (stretch)	c (colour)	μ
1993ah	0.0285	16.86 ± 0.19	-2.26 ± 0.93	0.23 ± 0.09	35.35 ± 0.22
1993ag	0.0500	17.79 ± 0.05	-1.09 ± 0.24	0.12 ± 0.02	36.68 ± 0.17
1993o	0.0529	17.60 ± 0.05	-1.03 ± 0.14	-0.01 ± 0.02	36.82 ± 0.16
1993b	0.0701	18.43 ± 0.04	-0.53 ± 0.21	0.09 ± 0.02	37.45 ± 0.16
1992bs	0.0627	18.25 ± 0.05	-0.27 ± 0.23	0.02 ± 0.02	37.48 ± 0.16
1992br	0.0876	19.19 ± 0.11	-2.97 ± 0.38	-0.04 ± 0.07	38.23 ± 0.19

TABLE VII: Example rows from the Union 2.1 Type Ia supernova dataset showing redshift z , peak B-band magnitude m_B , stretch parameter x_1 , colour parameter c , and distance modulus μ . The full compilation contains 714 supernovae spanning $0.0025 \leq z \leq 1.4140$ [5].

SCIENTIFIC SUMMARY FOR A GENERAL AUDIENCE

Astronomers study the Universe's expansion by measuring how far away objects are and how fast they are moving. One of the most useful "cosmic rulers" is a type of exploding star called a Type Ia supernova. These stars shine with a similar brightness, allowing scientists to estimate their distance by observing how bright they appear from Earth.

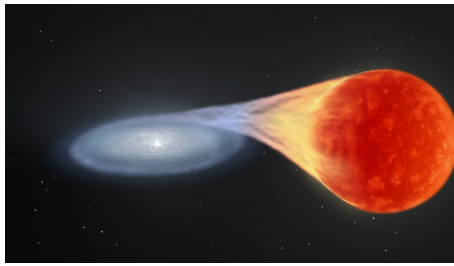


FIG. 13: Hypothetical Type Ia supernova forming by gaining mass from a neighbouring star until it explodes. Credit to NASA's Goddard Space Flight Center Conceptual Image Lab.

This project analysed observations of 714 supernovae, some of whose light has travelled for billions of years before reaching us. By comparing the observed brightness with theoretical predictions, we can estimate how much matter and dark energy (a mysterious quantity) fill the Universe. The balance of these components determines the history and possible fate of the Universe.

The results strongly support a Universe dominated by dark energy, consistent with previous studies. However, allowing for the possibility that the Universe is curved rather than flat reduced the precision of these measurements. This occurs because different combinations of the parameters can produce very similar observational results, a phenomenon known as degeneracy. Observations of more distant supernovae, together with other types of astronomical data, will help break this degeneracy and improve the confidence of future measurements.